PROJECT REPORT

Design and Modeling of Bottle and Threaded Cap Using Fusion 360

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1. Introduction

This project focuses on the design and 3D modeling of a plastic bottle and its corresponding threaded cap using Autodesk Fusion 360. The objective of this project was to understand parametric modeling, revolve features, thread creation, assembly design, and component management within a CAD environment.

The bottle and cap were modeled as separate components and later assembled to verify proper fitting and alignment. This project demonstrates practical product design skills used in packaging industries such as beverage, pharmaceutical, cosmetic, and household product manufacturing.

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2. Objectives

The main objectives of this project were:

- To design a parametric 3D model of a bottle using the revolve feature.
- To create an internally threaded cap that matches the bottle neck.
- To apply realistic dimensions and proportions.
- To assemble both components and verify functional compatibility.
- To understand Fusion 360's component structure and browser management.

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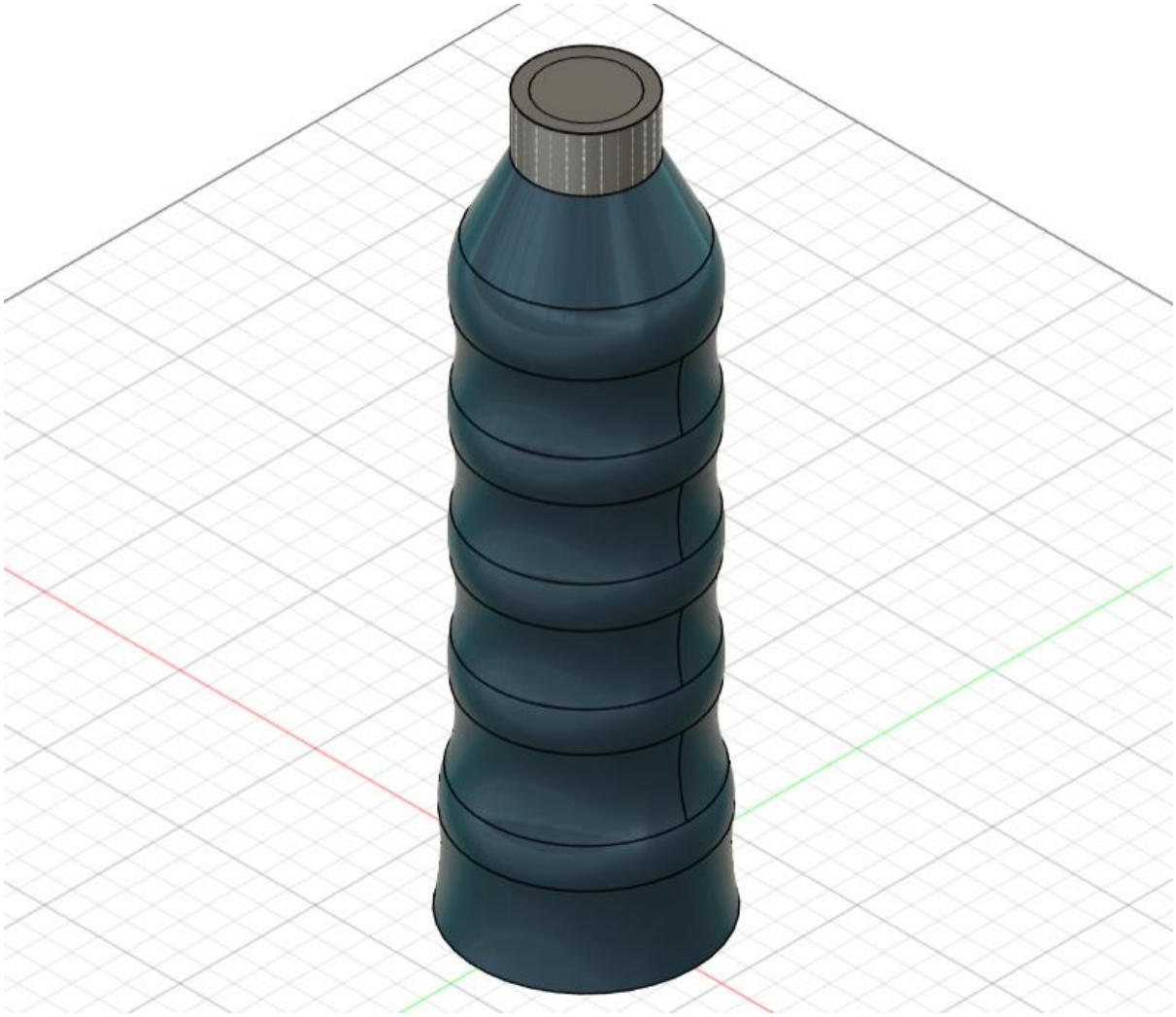
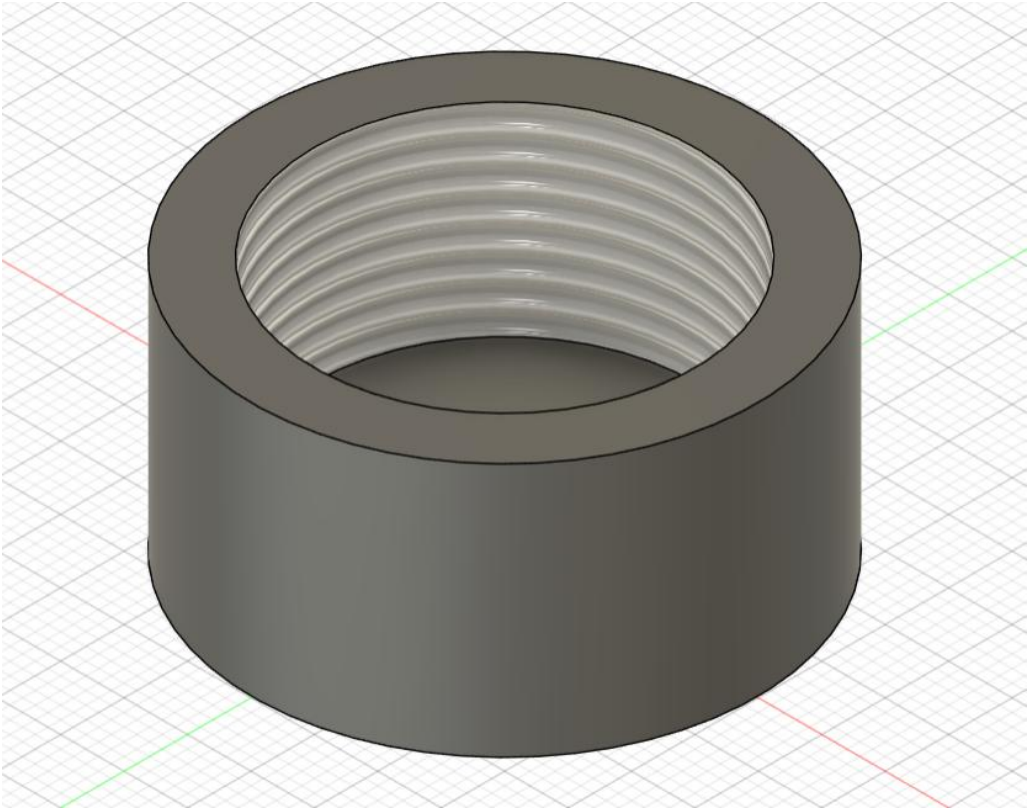
3. Software Used

Autodesk Fusion 360

Fusion 360 was selected because it provides:

- Parametric modeling capabilities
- Advanced solid modeling tools
- Thread creation tools
- Assembly design environment
- User-friendly interface with component browser





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4. Design Methodology

4.1 Bottle Design

The bottle body was created using the **Revolve feature** in Fusion 360.

Steps followed:

1. A 2D profile of half the bottle was sketched on a vertical plane.
2. The profile included:
    - Base diameter
    - Curved body grip sections
    - Tapered shoulder
    - Cylindrical neck
3. The sketch was fully constrained using dimensions.
4. The profile was revolved 360° around the center axis.
5. Shell command (if applied) was used to create hollow thickness.
6. External thread was added on the bottle neck.

Design Features of Bottle:

- Ergonomic curved grip sections
- Stable base support
- Tapered shoulder for aesthetic appeal
- Cylindrical threaded neck for cap fitting

4.2 Cap Design

The cap was designed as a separate component.

Steps followed:

1. A circle was sketched representing the outer diameter.
2. The profile was extruded to required height.
3. Shell command used to hollow the interior.
4. Internal threads were created to match bottle neck threads.
5. Proper thread pitch and diameter were aligned for accurate fitting.

Cap Features:

- Cylindrical outer body
- Internal threading

- Sufficient wall thickness
- Flat top surface

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5. Assembly and Component Structure

The bottle and cap were created as separate components under the main project file.

From the Fusion 360 browser structure:

- Main Project Folder
- Bottle Component
- Cap Component
- Bodies
- Sketches
- Joints

A joint was applied to align the cap with the bottle neck. The alignment verified correct thread compatibility and dimensional accuracy.

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6. Design Considerations

Several engineering considerations were taken into account:

6.1 Dimensional Accuracy

Proper matching of:

- Bottle neck diameter
- Cap internal diameter
- Thread pitch
- Clearance tolerance

6.2 Manufacturing Feasibility

The design is suitable for:

- Injection molding process
- Blow molding for bottle
- Plastic cap manufacturing

6.3 Ergonomics

- Curved body enhances grip
- Proper height-to-diameter ratio

- Balanced center of gravity

6.4 Structural Stability

- Thick base for support
- Uniform wall thickness
- Thread depth sufficient for sealing

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7. Working Principle

The working principle of this bottle and cap system is based on a **threaded fastening mechanism**.

When the cap is rotated clockwise:

- Internal threads of cap engage with external threads of bottle neck.
- Axial force is generated.
- Cap moves downward along the thread path.
- A tight seal is formed at the top surface.

When rotated counter-clockwise:

- Threads disengage.
- Cap moves upward.
- Bottle opens.

This mechanism ensures:

- Secure sealing
- Reusability
- Ease of operation

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8. Applications

This type of bottle-cap design is commonly used in:

- Beverage packaging
- Water bottles
- Pharmaceutical containers
- Cosmetic packaging
- Household liquid containers

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9. Challenges Faced

- Achieving perfect thread alignment
- Maintaining correct tolerance between bottle and cap
- Designing smooth curved body sections
- Managing separate components in assembly

These challenges were resolved by adjusting parametric dimensions and using Fusion 360's thread and joint tools effectively.

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10. Learning Outcomes

Through this project, the following skills were developed:

- Sketching and constraining profiles
- Revolve feature usage
- Shell operation
- Thread creation (internal & external)
- Assembly joints
- Component management
- Product design thinking

This project improved understanding of real-world product modeling and functional assembly validation.

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11. Conclusion

The bottle and cap model designed in Fusion 360 successfully demonstrates parametric modeling, threaded assembly design, and component management. The cap fits accurately onto the bottle neck, confirming correct dimensional planning and thread alignment.

The project reflects practical industrial product design methodology and enhances understanding of packaging engineering principles. It provides a strong foundation for advanced mechanical design and manufacturing applications.